

1234

LOGGY SENSOR BOARD: ENGG3800 TEAM6 2025 Sem2

A

MCU, LCD CONTROL AND SD CARD.+3.3V SOURCE, +5V SOURCE (SEEDUINO) CONTROL TOSLINK TRANSCEIVER

SS1
MCU_LCD_SD_PROG.SchDoc



A Pinout and communication ports on U1:STM32L433CCT6 for LCD and SD card peripherals as well as user interface. Programming header and coin battery supply also.

SS2
VOLTAGECURRENT.SchDoc



A Board is powered from SEEDUINO +5V output. +5V source is switch controlled to board, supplies +3.3V rail linear regulator.

SS3
TOSLINK.SchDoc



A Utilises U5: MAX941CPA+ to communicate with sensor board via TOSLINK cable. UART protocol with throughhole transmit LED and BPW34 recieve photodiode.

UI PUSHBUTTONS AND MECHANICALS

SS4
UI&MECH.SchDoc



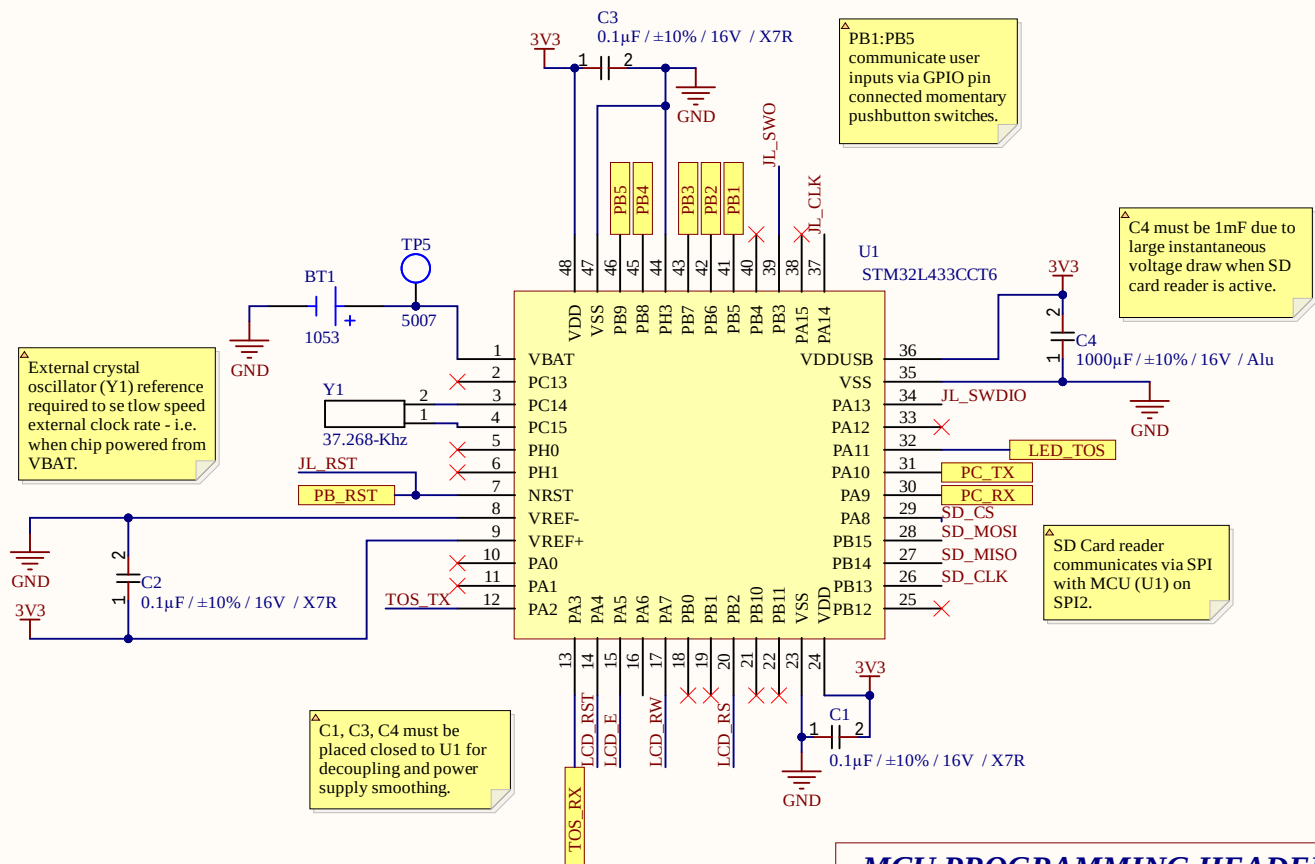
A Contains mechanical components and pushbuttons for user interface control. Also includes mounting hardware.

D

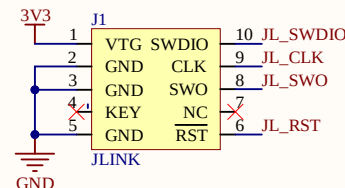
Title			CONTROL BOARD TITLE PAGE AND SCHEMATIC SYMBOLS
Size	Number	Revision	
A4	1	1	
Date:	5/26/2025	Sheet0 of 4	
File:	X:\Files\.\LANDING_PAGE.SchDoc	Drawn By: TEAM06	

1234

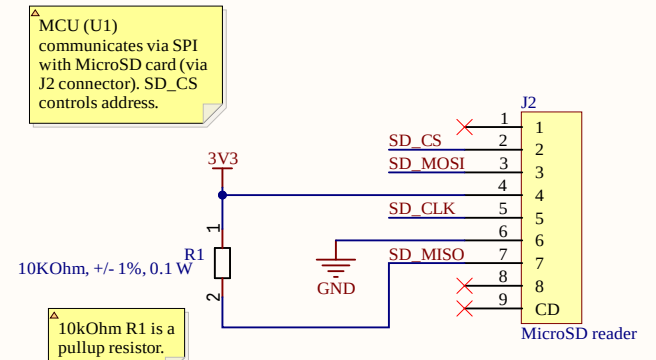
MCU PINOUT, COMMUNICATION PORTS AND PROGRAMMING.



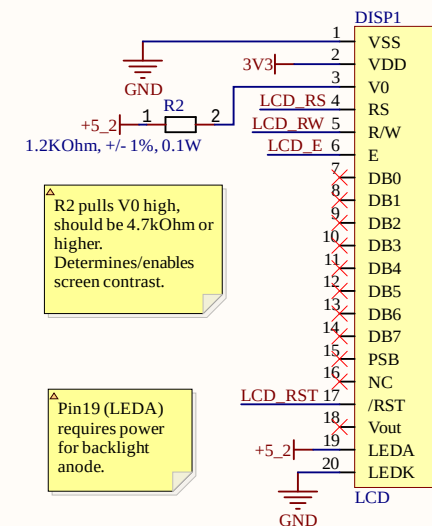
MCU PROGRAMMING HEADER: SWD



MICRO SD CARD INTERFACE

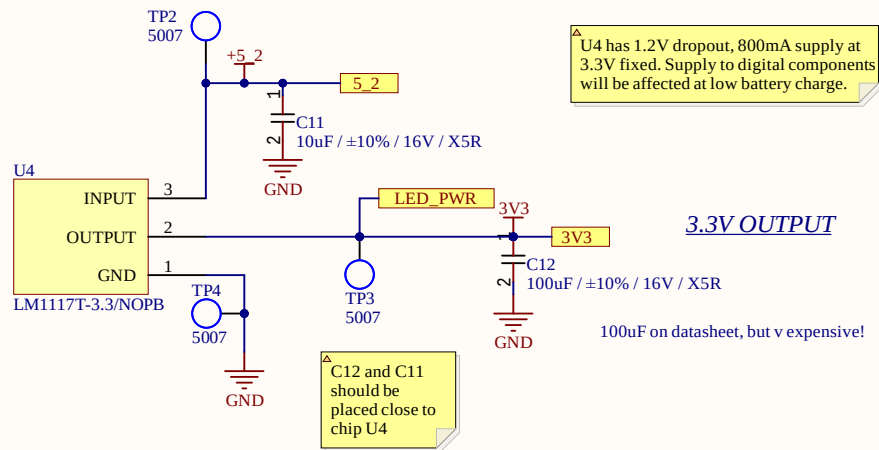


LCD SCREEN: USER DISPLAY



Title		
MCU, LCD and SD CARD		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet 1 of 4
File:	X:\Files\...MCU_LCD_SD_PROG_SchDoc	Drawn By: TEAM06

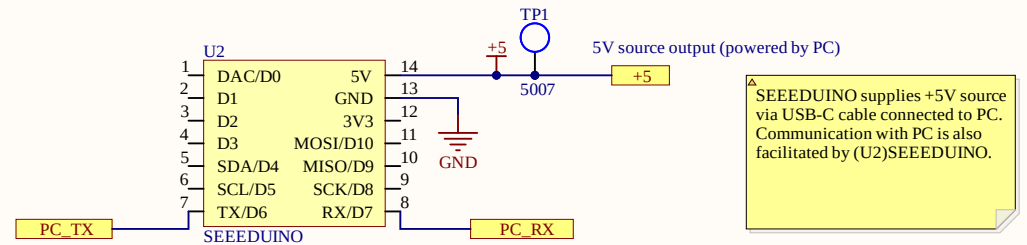
3.3V SOURCE



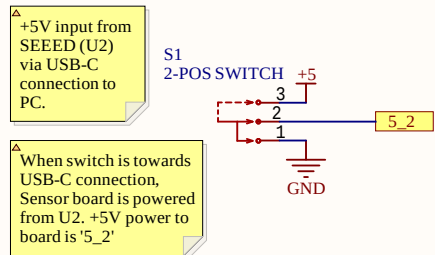
3.3V OUTPUT

SEEDUINO

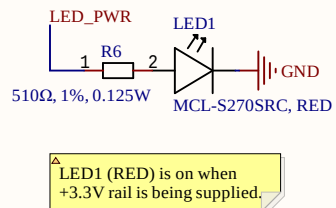
5V source and PC communication



5V POWER SWITCH

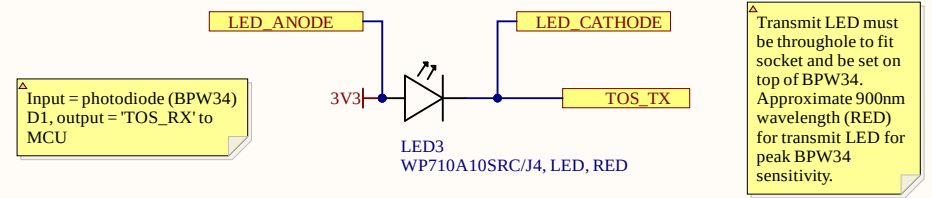


POWER LED

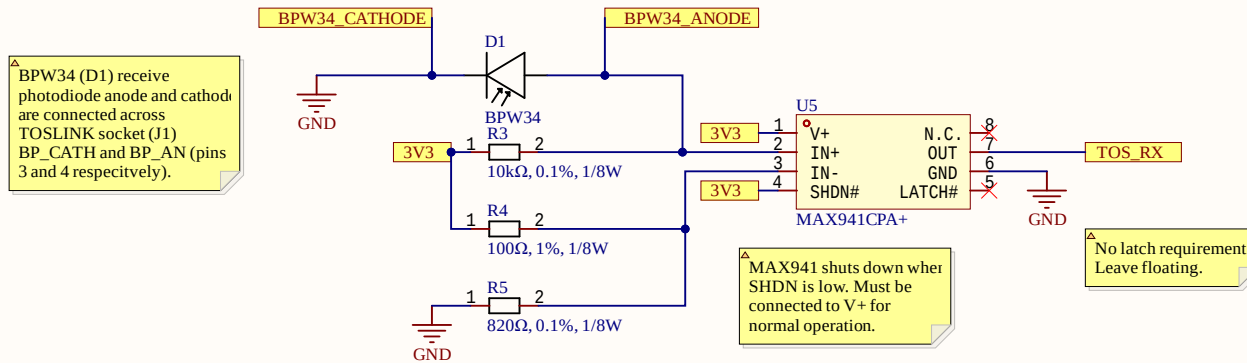


Title		
VOLTAGE SOURCES AND POWER CONTROL		
Size	Number	Revision
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Date:	5/26/2025	Sheet2 of 4
File:	X:\Files\...\VOLTAGECURRENT.SchDoc	Drawn By: TEAM06

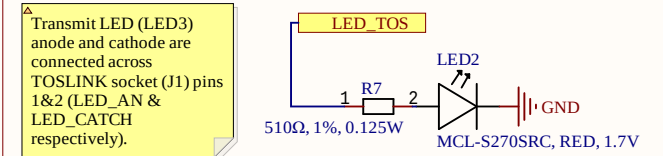
TOSLINK TRANSMIT



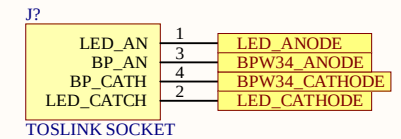
TOSLINK RECEIVE



SAMPLING LED



TOSLINK SOCKET CONNECTIONS



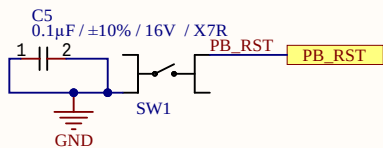
TOSLINK socket is mounted around LED3 (WP710A10SRC/J4) and D1 (BPW34). LED3 must be mounted across BPW34 photosensitive face. Lead holes for each diode are orthogonal to each other.

Title		
TOSLINK TRASCEIVER AND SAMPLING LED		
Size	Number	Revision
A4	1	1
Date:	5/26/2025	Sheet3 of 4
File:	X:\Files\...\TOSLINK.SchDoc	Drawn By: TEAM06

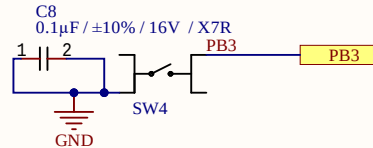
USER INTERFACE - PUSHBUTTONS

6 momentary pushbuttons toggle user selections and control. Capacitors C5-C10 for transient mitigation.

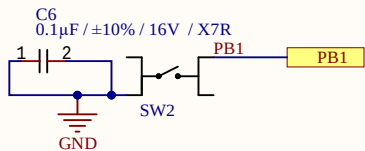
RESET PUSHBUTTON



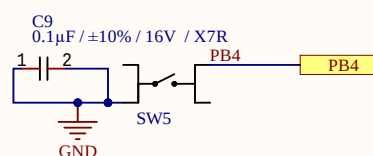
CHANNEL SELECT PUSHBUTTON



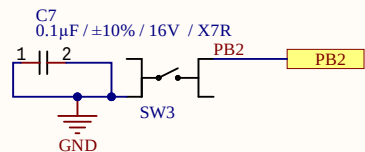
NEXT PAGE PUSHBUTTON



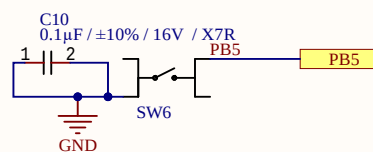
ALARM STATUS PUSHBUTTON



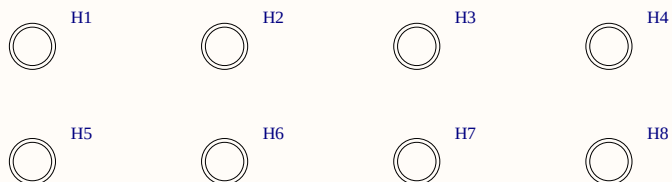
RANGE SELECT PUSHBUTTON



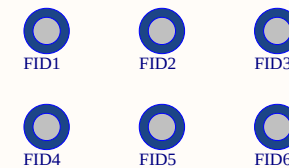
SD CARD PUSHBUTTON



MOUNTING HOLES

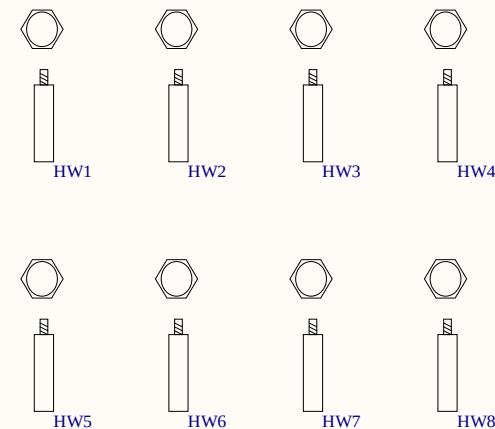


FIDUCIALS



STANDOFFS

Standoffs comprised of pillar and nut



Title USER INTERFACE PUSHBUTTONS AND PERIPHERALS		
Size A4	Number 1	Revision 1
Date: 5/26/2025	Sheet 4 of 4	
File: X:\Files\UI&MECH.SchDoc	Drawn By: TEAM06	